

APPLIED DATA SCIENCE CAPSTONE PROJECT

SpaceX Falcon 9 First-Stage Landing Prediction

Jeffrey Alwin

Machine learning · geospatial analytics · REST APIs · SQL · data visualization.

https://github.com/Jeffrey-Alwin/Capstone

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THE QUESTION

Landing success is predictable from mission context

Objective

Predict first-stage landing success to evaluate reusable-launch economics.

Pipeline

API and web data → wrangling → SQL and EDA → maps and dashboard → four classifiers.

Result

All four tuned models reached **83.33% test accuracy.**

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WHY RECOVERY MATTERS

Reusable boosters change the economics of launch

Falcon 9 recovery creates a reusable operating model where traditional launches discard expensive rocket stages.

Prediction problem

Estimate landing feasibility before launch from observable mission parameters.

Strategic value

Inform pricing, risk assessment, and launch-contracting decisions.

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DATA FOUNDATION

Two public data sources built the mission dataset

After filtering, the project isolated **90 operational Falcon 9 launches.**

SpaceX REST API

Paginated launch records enriched through rockets, payloads, launchpads, and cores endpoints.

Wikipedia tables

Flight number, date, booster version, site, payload, orbit, and customer metadata.

Connected ingestion pipeline

Two sources | Enrichment joins | Final launch table.

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DATA FOUNDATION

Wikipedia tables complete the launch history

A structured web-ingestion path fills the historical gaps around the REST API dataset.

HTTP GET Request

HTML DOM Tree

BeautifulSoup Parsing

Row Extraction

Clean Pandas DataFrame

HTTP GET: requests fetch the raw Wikipedia HTML. **Parse:** BeautifulSoup navigates the nested DOM with `html.parser`. **Target:** select `wikitable plainrowheaders collapsible` tables.

Extract: rows and cells yield flight, booster, site, payload, orbit, customer, and outcome fields. **Clean:** remove citation markers and line breaks before assembling one Pandas DataFrame.

Outcome

Complete historical flight records complement and validate the REST API dataset.

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MODEL INPUTS

Feature engineering translated recovery outcomes into model inputs

01

Clean

Missing payload values were filled with central-tendency estimates.

02

Define

Class = 1: successful RTLS, ASDS, or ocean recovery.
Class = 0: failed or absent recovery.

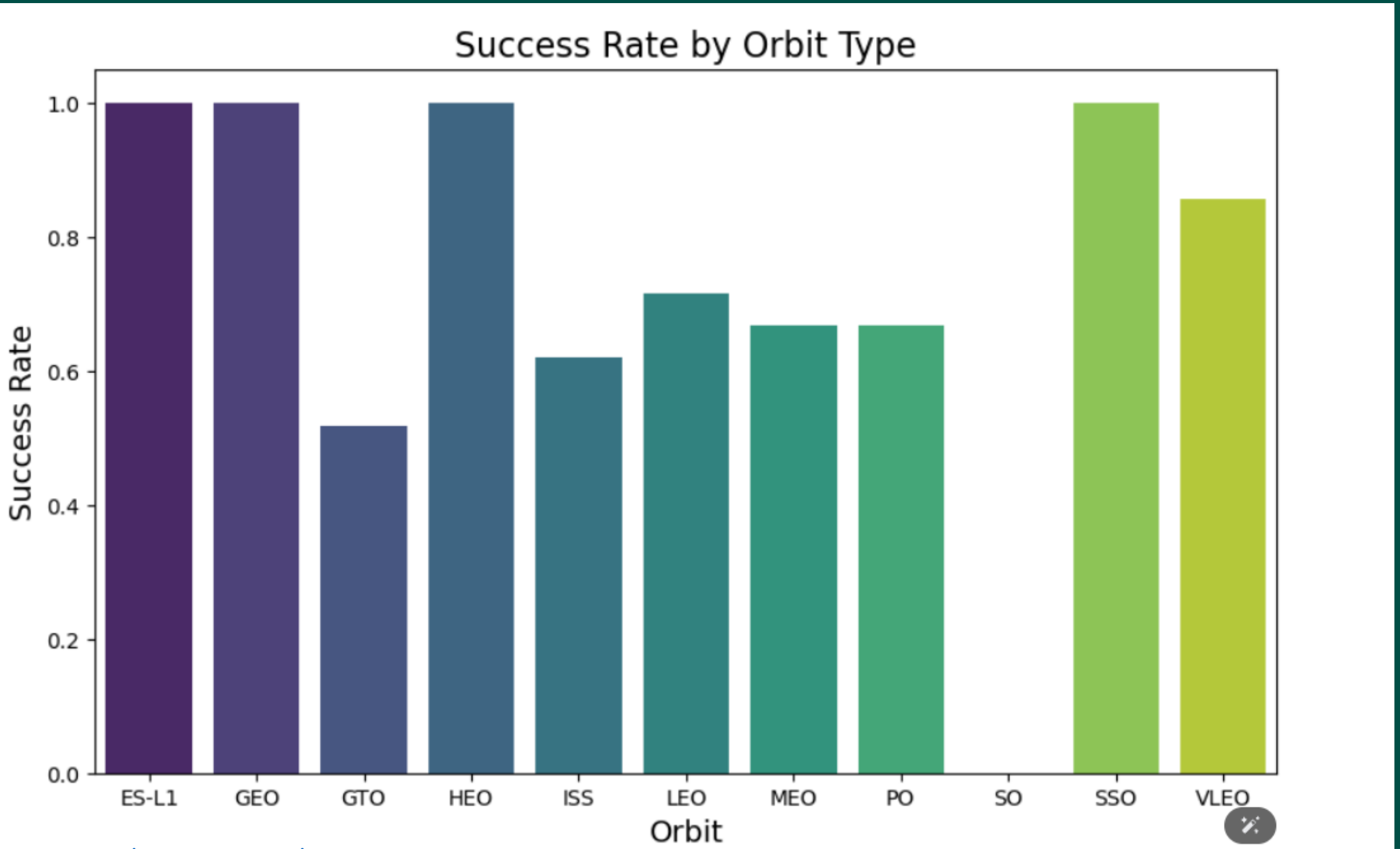
03

Expand

One-hot encoding expanded orbit, site, landing pad, and booster serial into **83 predictors.**

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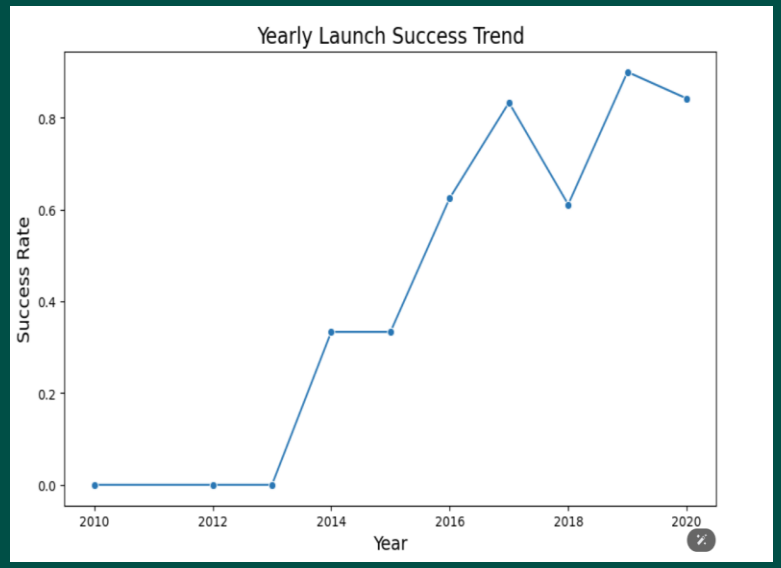
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EXPLORATORY ANALYSIS

Launch history reveals where recovery became repeatable

YEARLY TREND LINE

0% → >80%



SITE EXPANSION

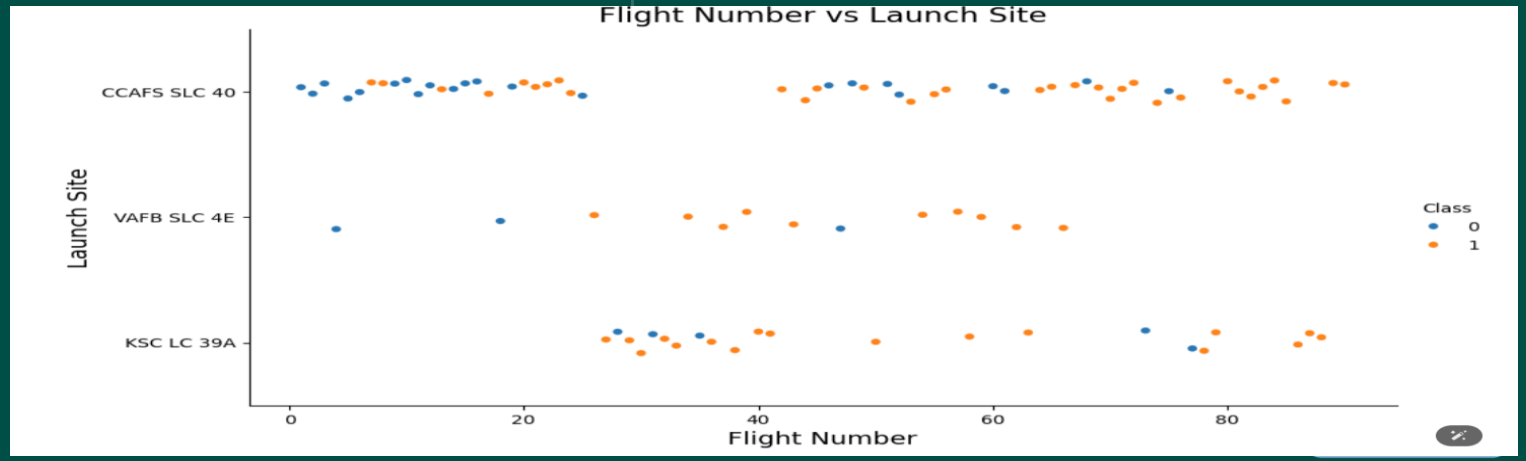
CCAFS SLC-40
VAFB SLC-4E
KSC LC-39A

Later missions show operational expansion across three launch sites.

PAYLOAD × SITE

KSC + CCAFS

Heavy payloads cluster at KSC and CCAFS; VAFB polar launches tend to carry lighter payloads.



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QUERYABLE HISTORY

SQL made the mission history queryable

Representative findings from the launch database.

Query	Finding
Distinct launch sites	**4 sites**
NASA payload aggregation	**45,596 kg**
F9 v1.1 average payload	**2,928.4 kg**
Earliest ground landing	**2015-12-22**
Drone-ship payload range	**4,000–6,000 kg**

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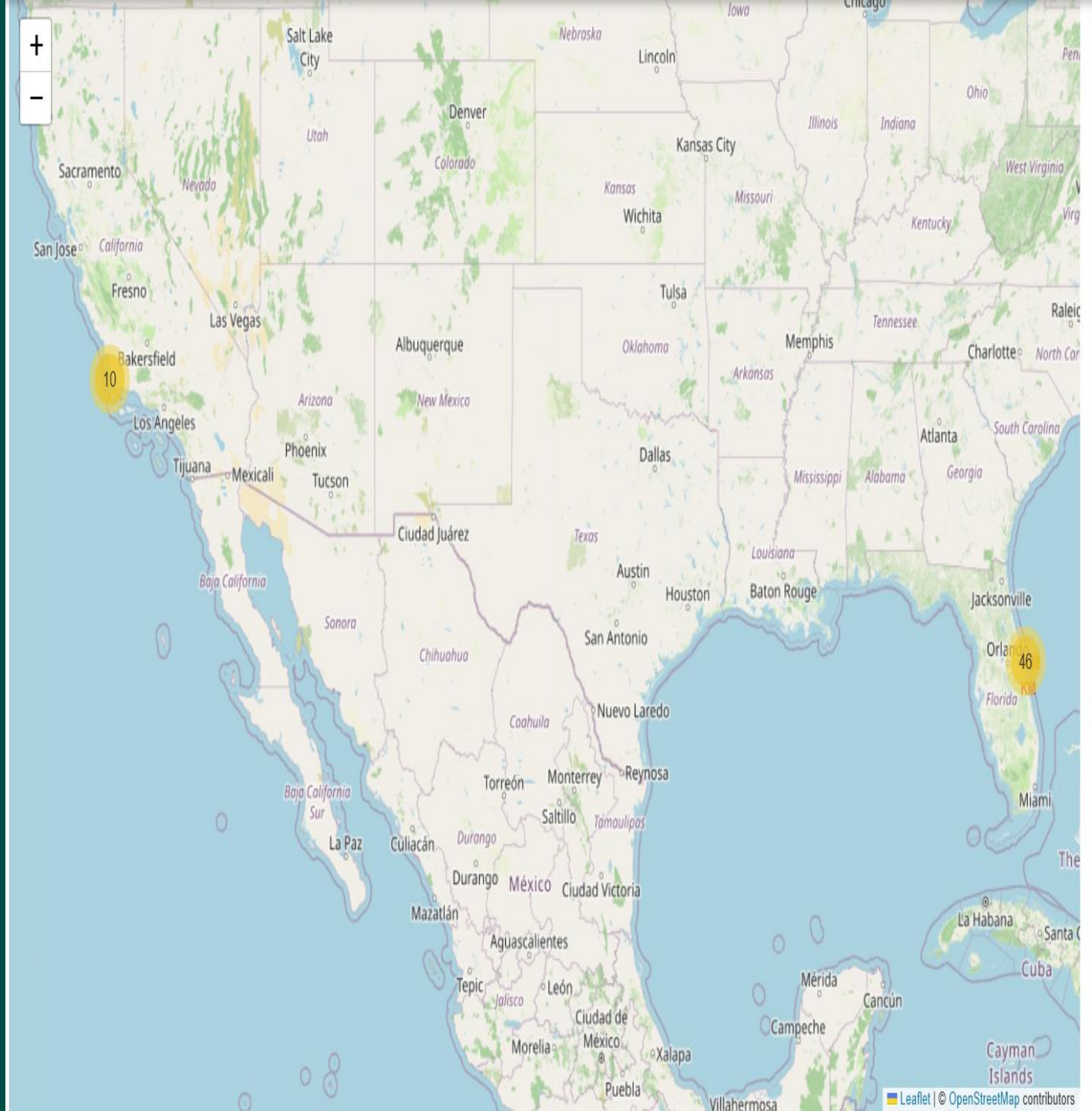
GEOSPATIAL ANALYTICS

Geospatial context explains recovery logistics

Folium mapped launch coordinates and colored outcomes by success or failure. Marker clusters kept dense launch histories legible across zoom levels.

SUCCESS FAILURE

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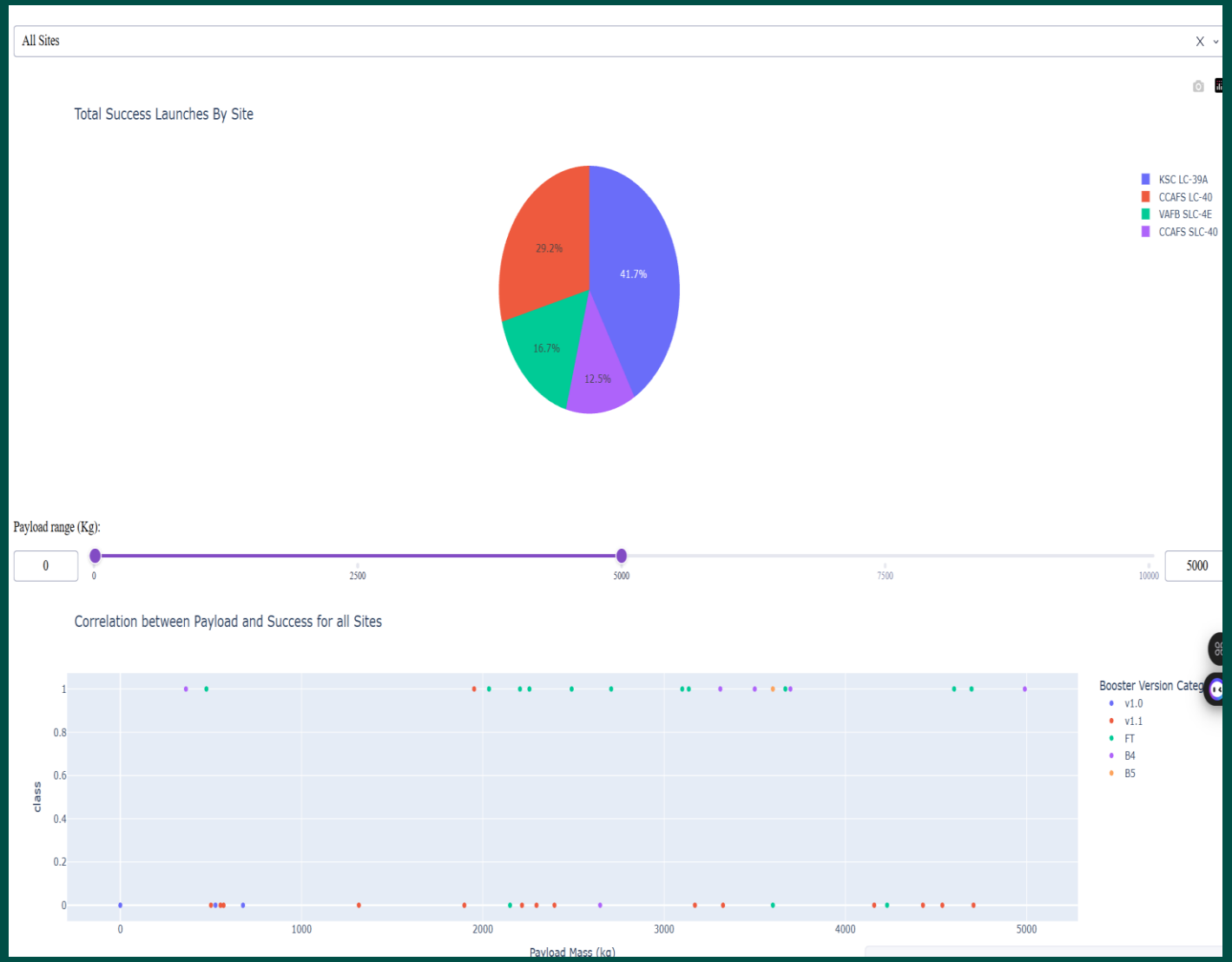
DECISION SURFACE

The dashboard turns the dataset into an interactive decision surface

Users filter by launch site and payload range from 0 to 10,000 kg.

Reactive outputs

Success versus failure proportions, plus payload mass, outcome, and booster version.



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MODEL COMPARISON

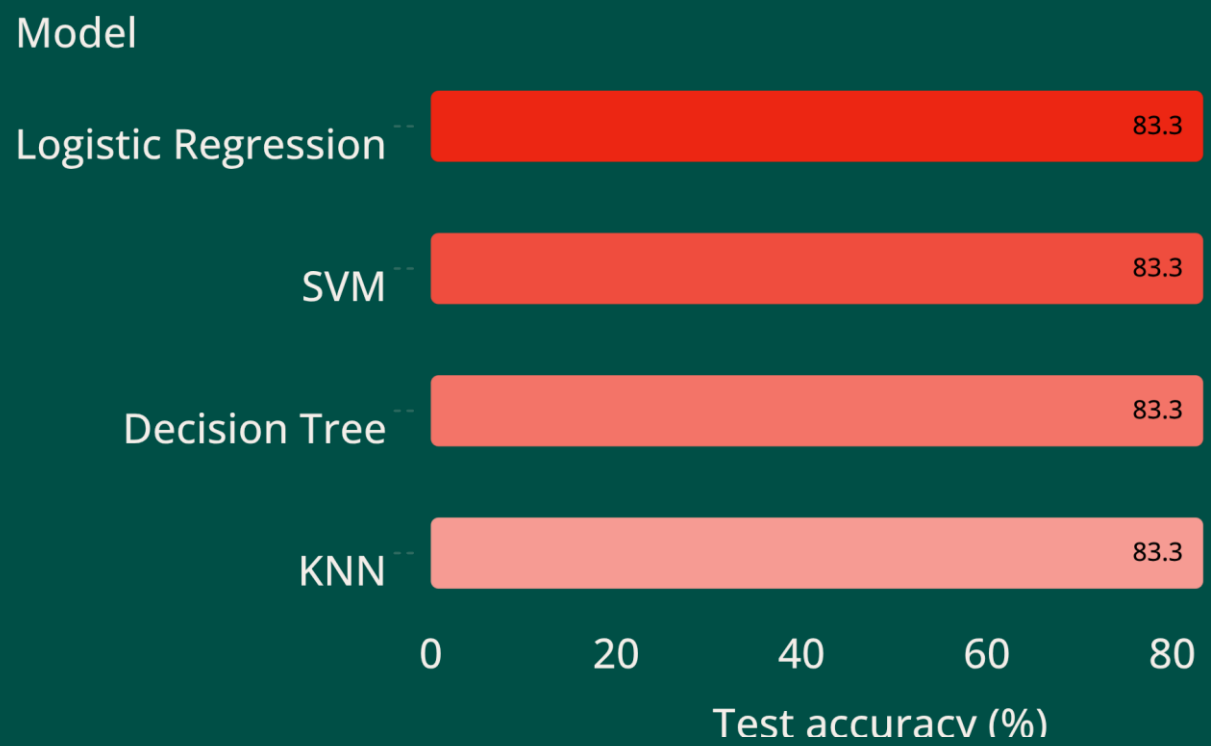
Four tuned classifiers converged on the same accuracy

Features were standardized and split 80/20 with `random_state=2`.

GridSearchCV used 10-fold cross-validation across Logistic Regression, SVM, Decision Tree, and KNN.

83.33%

Test accuracy for every tuned model.



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MODEL EVALUATION · TEST SET

Predictive Analysis Results & Confusion Matrix

Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors each achieved 83.33% test accuracy through 10-fold GridSearchCV cross-validation.

12	3	3	0
True Positives	True Negatives	False Positives	False Negatives

2x2 confusion matrix heatmap

Actual vs. predicted: TP 12 · FN 0 · FP 3 · TN 3

Test-accuracy comparison bar chart

LR · SVM · Decision Tree · KNN: 83.33% each

BEST MODEL VERDICT

Decision Tree Classifier for production

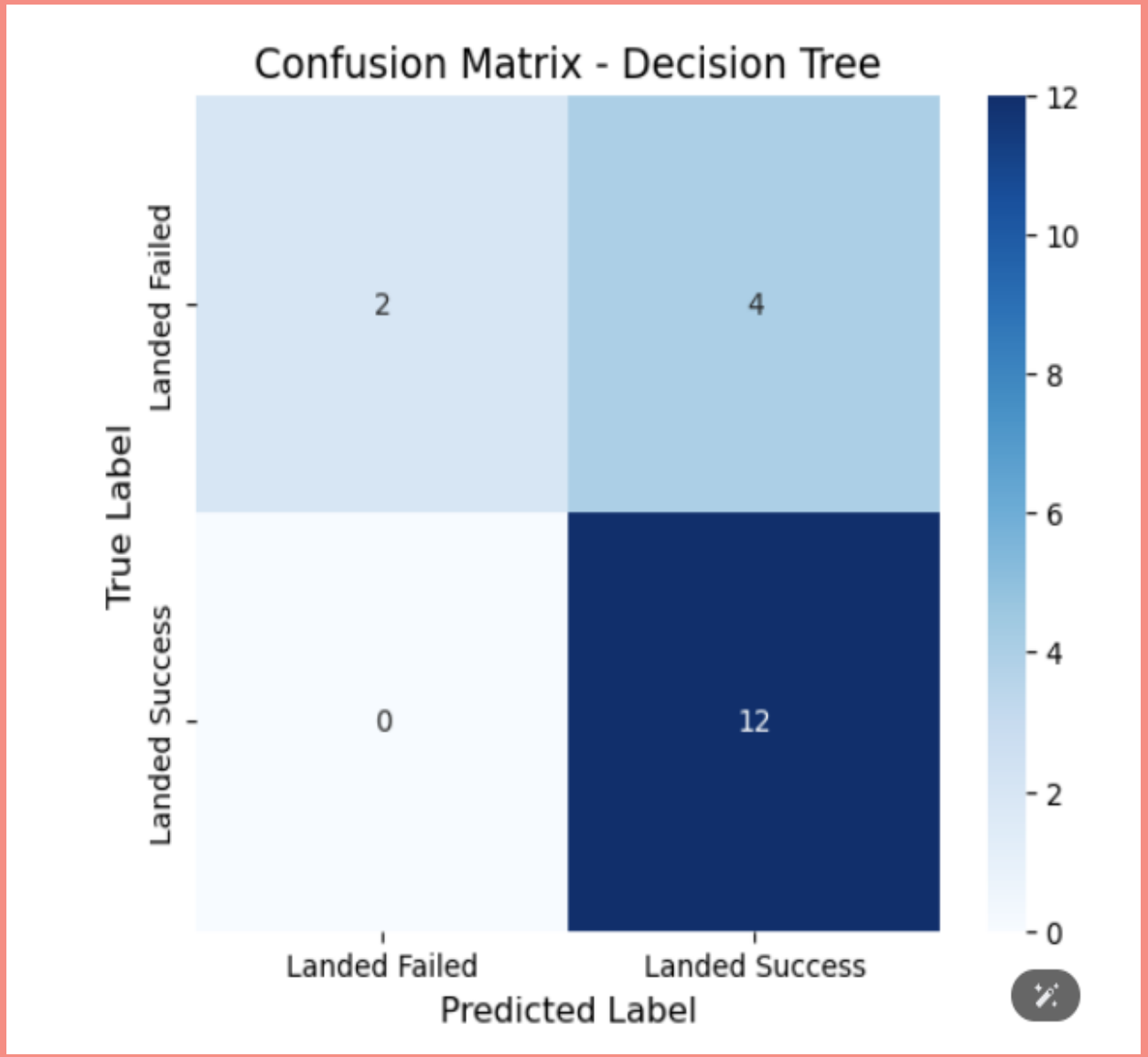
Explicit, rule-based decision paths provide operational transparency for aerospace stakeholders.

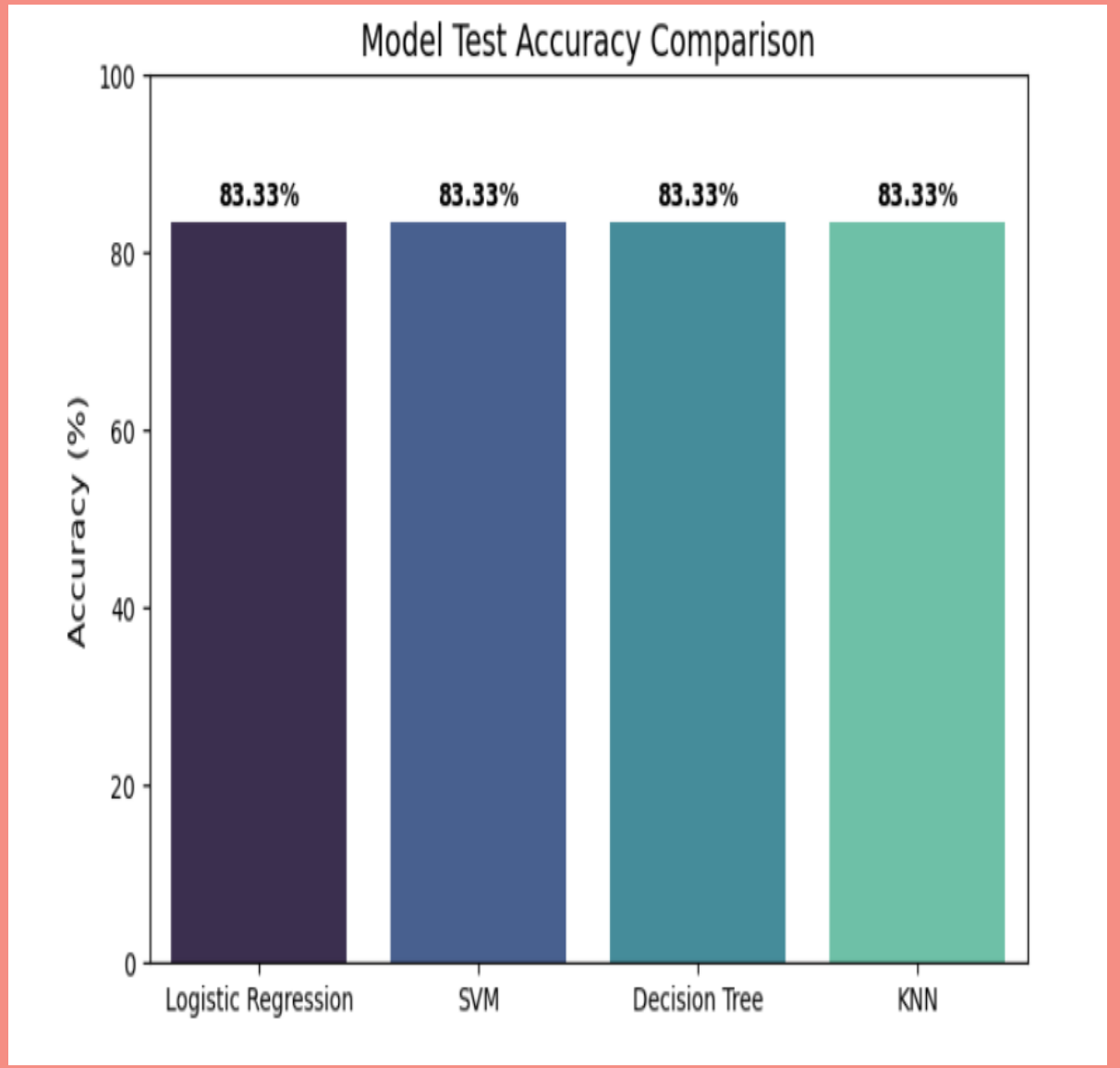
Recall: 100% · **Precision:** ~80%

Zero landing successes misclassified as crashes.

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CONCLUSION

Conclusion & Innovative Insights

Core Operational Conclusions

- **Payload mass** governs landing feasibility: heavier ASDS drone-ship missions carry narrower fuel margins.
- **Orbital destination:** high-energy GTO missions impose higher thermal and velocity stresses than LEO.
- **Launch site** location is a key factor in recovery outcomes.
- **Trajectory:** recovery success rose from 0% in 2010–2012 to consistently above 80% after 2017.

Creative & Innovative Insights

- **Competitive pricing:** commercial satellite competitors can use the classification engine to benchmark launch pricing against SpaceX's $62M cost baseline.
- **Insurance underwriting:** aerospace underwriters can calculate predictive risk premiums from quantifiable pre-launch flight parameters.

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